

Software Installation Manual (Metal count App)

Windows Installation Manual

Mosquitto –MySQL–Python Metal Counting Application

1. Introduction

This guide explains how to install and configure the Metal Counting Application on **Windows 10/11**, using:

Python 3.10 – application logic

MQTT (Eclipse Mosquitto) – sensor communication

MySQL Community Server – data storage

2. System Requirements

CPU: Dual-core 1.5 GHz or higher

RAM: 4 GB minimum (8 GB recommended)

Storage: 10 GB free

Network: Ethernet or Wi-Fi with MQTT broker access

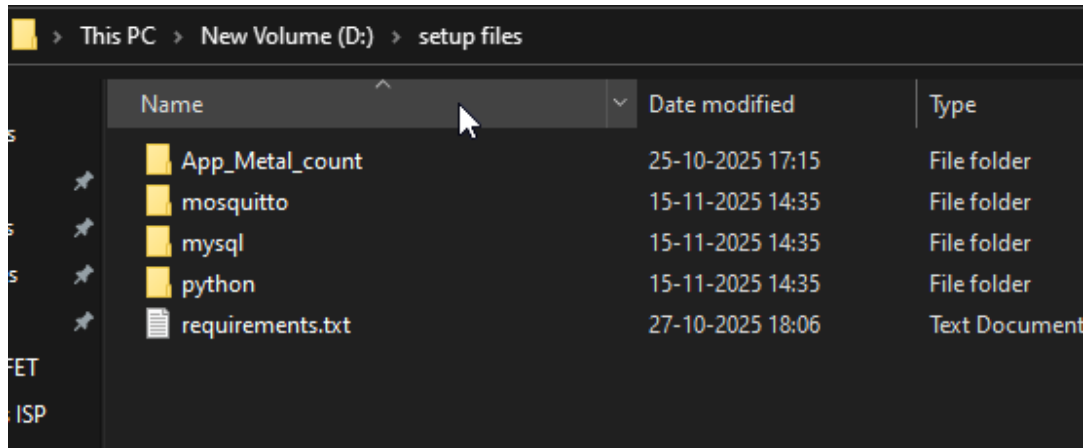
3. Software Components Overview

Component	Purpose
Python 3.10	Core environment for the app
paho-mqtt	MQTT client library
MySQL Server	Database for counts and logs

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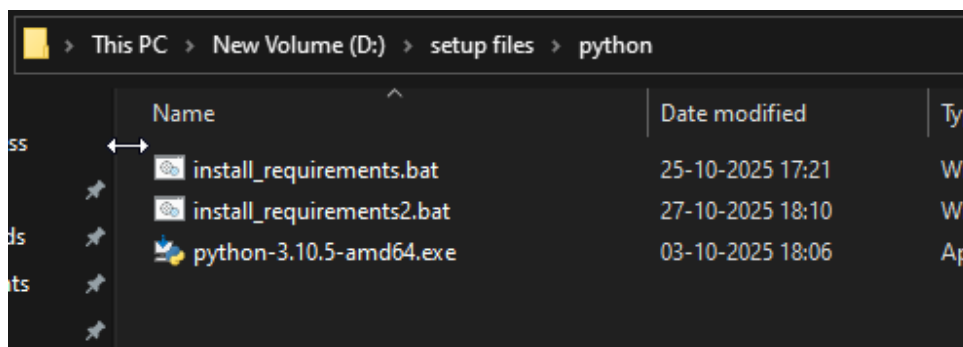
4. Installation Steps

“setup files” folder has all the installation files.



Directory Structure

4.1 Install Python



Download Python 3.10 from <https://www.python.org/downloads/windows/>

Run installer:

Check “Add Python to PATH”

Select “Install for all users”

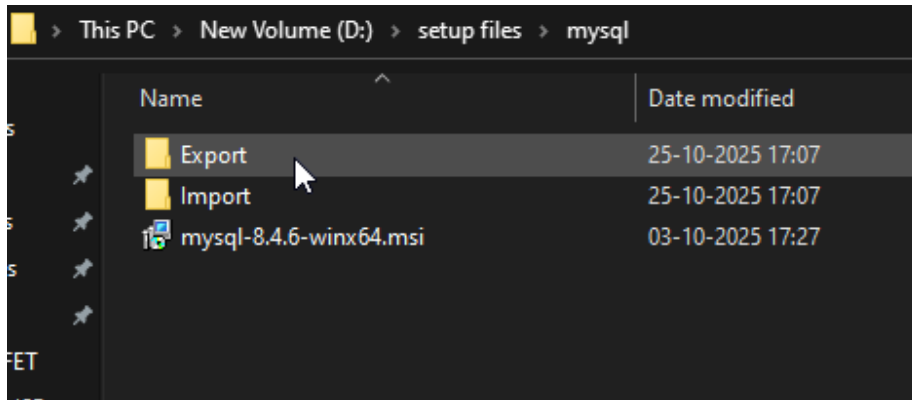
Verify installation:

Run in Cmd “python –version”

Run in Cmd “pip –version”

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4.2 Install MySQL Community Server



Download installer from <https://dev.mysql.com/downloads/installer/>

Run installer:

Choose “Developer Default”

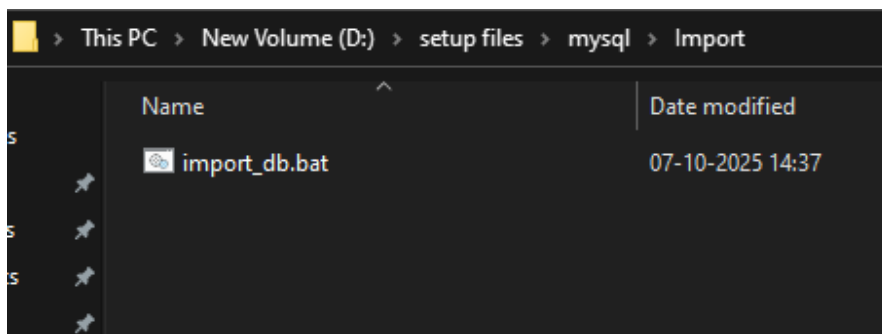
Set a root password (remember it!)

Configure MySQL as a Windows service (default)

Verify installation:

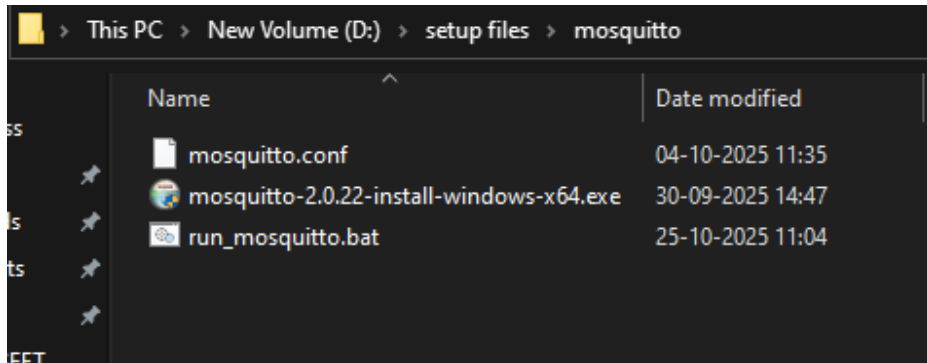
Run in Cmd “mysql -u root -p”

Import device_db by double clicking \Import\ import_db.bat



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4.3 Install MQTT Broker (Mosquitto)



Download Mosquitto for Windows: <https://mosquitto.org/download/>

Install with default options. Check “Install service” to run Mosquitto as Windows service.

Verify service is running:

Run in Cmd “net start | findstr Mosquitto”

Optional: Enable username/password authentication by editing:

C:\Program Files\mosquitto\mosquitto.conf

allow_anonymous false

password_file C:\Program Files\mosquitto\passwd.txt

Create user:

mosquitto_passwd -c "C:\Program Files\mosquitto\passwd.txt" mqttuser

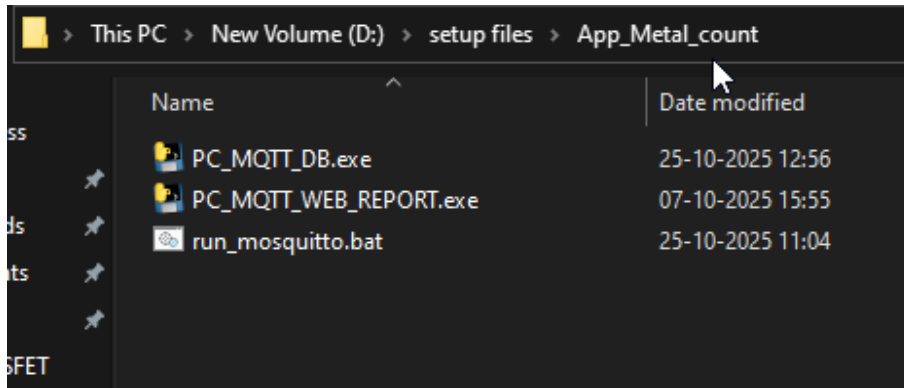
Restart Mosquitto service:

Run in Cmd “net stop mosquito”

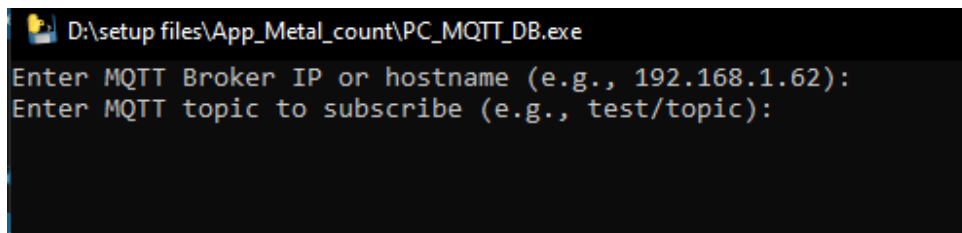
Run in Cmd “net start mosquito”

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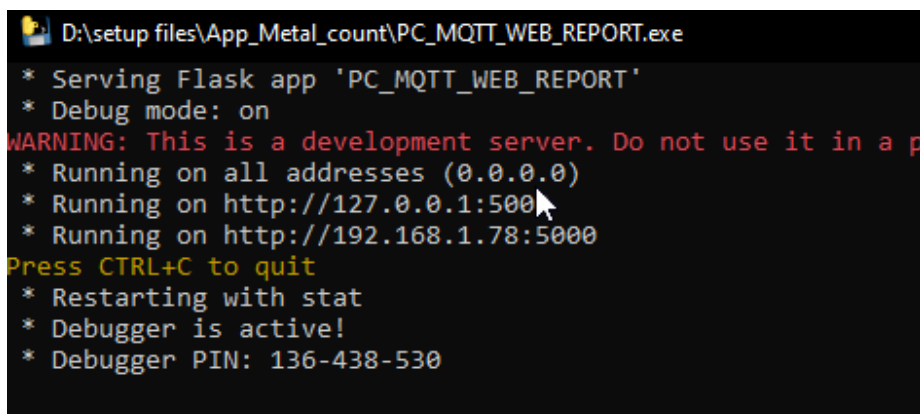
4.4 Run Applications



Step 1: double click PC_MQTT_DB.exe. It will asks for mqtt broker IP address, mqtt password.

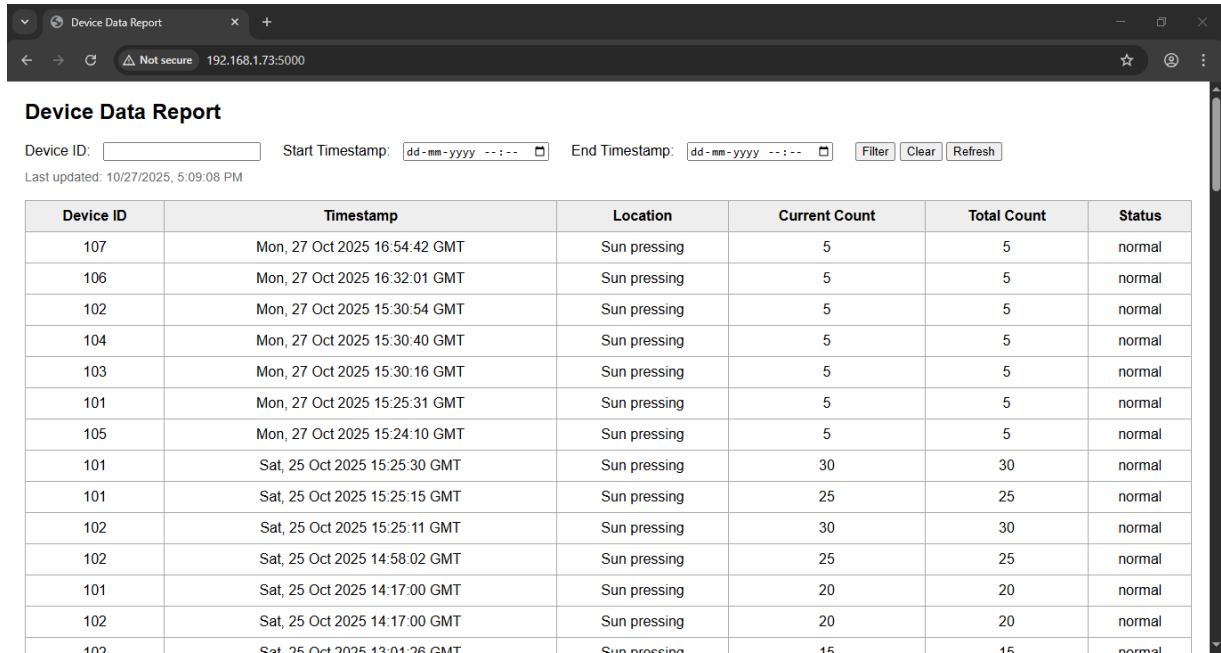


Step 2: double click PC_MQTT_WEB_REPORT.exe. It will show the report IP address.



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Copy and paste the IP address in browser and press enter to view report.



The screenshot shows a web browser window with the address bar displaying '192.168.1.73:5000'. The page title is 'Device Data Report'. Below the title, there are input fields for 'Device ID', 'Start Timestamp' (with a date picker), and 'End Timestamp' (with a date picker). There are also buttons for 'Filter', 'Clear', and 'Refresh'. Below these fields, it says 'Last updated: 10/27/2025, 5:09:08 PM'. The main content is a table with 6 columns: Device ID, Timestamp, Location, Current Count, Total Count, and Status. The table contains 15 rows of data, showing device counts over time from October 25 to October 27, 2025. All status values are 'normal'.

Device ID	Timestamp	Location	Current Count	Total Count	Status
107	Mon, 27 Oct 2025 16:54:42 GMT	Sun pressing	5	5	normal
106	Mon, 27 Oct 2025 16:32:01 GMT	Sun pressing	5	5	normal
102	Mon, 27 Oct 2025 15:30:54 GMT	Sun pressing	5	5	normal
104	Mon, 27 Oct 2025 15:30:40 GMT	Sun pressing	5	5	normal
103	Mon, 27 Oct 2025 15:30:16 GMT	Sun pressing	5	5	normal
101	Mon, 27 Oct 2025 15:25:31 GMT	Sun pressing	5	5	normal
105	Mon, 27 Oct 2025 15:24:10 GMT	Sun pressing	5	5	normal
101	Sat, 25 Oct 2025 15:25:30 GMT	Sun pressing	30	30	normal
101	Sat, 25 Oct 2025 15:25:15 GMT	Sun pressing	25	25	normal
102	Sat, 25 Oct 2025 15:25:11 GMT	Sun pressing	30	30	normal
102	Sat, 25 Oct 2025 14:58:02 GMT	Sun pressing	25	25	normal
101	Sat, 25 Oct 2025 14:17:00 GMT	Sun pressing	20	20	normal
102	Sat, 25 Oct 2025 14:17:00 GMT	Sun pressing	20	20	normal
102	Sat, 25 Oct 2025 13:01:26 GMT	Sun pressing	15	15	normal

5. Troubleshooting Guide

Issue	Solution
MQTT connection fails	Check broker IP, port, username/password; test with <code>mosquitto_sub -t test -v</code>
MySQL connection fails	Confirm user privileges, DB exists, service is running
Python errors	Ensure virtual environment activated; reinstall dependencies

6. Verification Checklist

Task	Status
Python installed	✓
MySQL installed & DB created	✓
Mosquitto running	✓
Dependencies installed	✓
App connects to MQTT & MySQL	✓
Auto-start service works	✓

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